Machine Learning Algorithms Reference

Supervised Learning Algorithms

Supervised learning trains models on labeled data to learn a mapping from inputs to outputs [Bishop, 2006].

Algorithm	Type	Complexity	Interpretable	Best For
Linear Regression	Regression	$O(n \cdot d)$	High	Continuous output prediction
Logistic Regression	Classification	$O(n \cdot d)$	High	Binary classification
Decision Tree	Both	$O(n \cdot d \cdot \log n)$	High	Rule-based decisions
Random Forest	Both	$O(n \cdot d \cdot t)$	Medium	Tabular data, feature importance
Gradient Boosting	Both	$O(n \cdot d \cdot t)$	Low	High-accuracy tabular tasks
SVM	Both	$O(n^2 – n^3)$	Low	Small datasets, high-dimensional
Neural Network	Both	$O(n \cdot d \cdot l)$	Very Low	Images, text, sequences

n = samples, d = features, t = trees/estimators, l = layers

Unsupervised Learning Algorithms

Unsupervised learning discovers hidden structure in unlabeled data [Hastie et al., 2009].

Algorithm	Output	Scalability	Hyperparameters	Best For
K-Means	Clusters	High	k (num clusters)	Spherical, compact clusters
DBSCAN	Clusters	Medium	ε, min_samples	Arbitrary-shaped clusters
PCA	Components	High	n_components	Dimensionality reduction
t-SNE	2D/3D layout	Low	perplexity	Visualization
Autoencoder	Latent space	Medium	Architecture	Anomaly detection
LDA	Topics	Medium	num_topics	Topic modeling in text

Training Pipeline

A standard ML training pipeline consists of the following phases:

1. **Data Preparation**
   - Collection
     - Structured sources: databases, CSV files, APIs
     - Unstructured sources: web scraping, PDFs, images
   - Cleaning
     - Remove or impute missing values
     - Detect and handle outliers (IQR method or z-score > 3)

        - Deduplicate records
    - Splitting
        - Training: 70–80% of data
        - Validation: 10–15% of data
        - Test: 10–15% of data (held out until final evaluation)

2. **Feature Engineering**
    - Numerical: scaling (StandardScaler, MinMaxScaler), binning
    - Categorical: one-hot encoding, target encoding, label encoding
    - Text: TF-IDF, word embeddings, tokenization
    - Temporal: lag features, rolling statistics, seasonal decomposition
3. **Model Training**
    - Select algorithm based on problem type and data characteristics
    - Tune hyperparameters using cross-validation (k-fold, stratified k-fold)
    - Monitor training loss and validation loss for overfitting signs
4. **Evaluation**
    - Classification metrics: accuracy, precision, recall, F1-score, AUC-ROC
    - Regression metrics: MAE, RMSE, R^2
    - Always evaluate on the held-out test set

Evaluation Metrics for Classification

Metric	Formula	Range	Notes
Accuracy	(TP + TN) / Total	0–1	Misleading on imbalanced data
Precision	TP / (TP + FP)	0–1	Minimizes false positives
Recall	TP / (TP + FN)	0–1	Minimizes false negatives
F1-Score	2 · (P · R) / (P + R)	0–1	Harmonic mean of precision/recall
AUC-ROC	Area under the ROC curve	0.5–1	Threshold-independent

> ***Note:*** *For severely imbalanced datasets (e.g., fraud detection where fewer than 0.1% of cases are positive), prefer* ***AUC-ROC*** *or* ***F1-Score*** *over accuracy — accuracy can be misleadingly high when the model simply predicts the majority class every time [Japkowicz & Stephen, 2002].*

References

- [Bishop, 2006] C.M. Bishop, *Pattern Recognition and Machine Learning*, Springer, 2006.
- [Hastie et al., 2009] T. Hastie, R. Tibshirani, J. Friedman, *The Elements of Statistical Learning*, 2nd ed., Springer, 2009.
- [Japkowicz & Stephen, 2002] N. Japkowicz, S. Stephen, "The Class Imbalance Problem: A Systematic Study," *Intelligent Data Analysis*, 2002.